



BITS Pilani
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Introduction to Statistical Methods

ISM Team

➤ **Section: 5**

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Random Variables

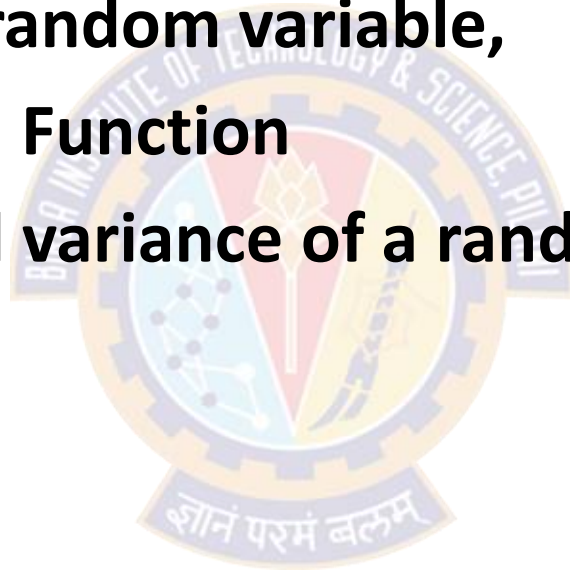
Contact Session 5

Contact Session 5: Module 3: Probability Distributions

Contact Session	List of Topic Title	Reference
CS - 5	Random variables - Discrete & continuous Expectation of a random variable, mean and variance of a random variable – Single random variable & Joint distributions	T1 & T2
HW	Problems on random variables	T1 & T2
Lab	Probability Distributions & Sampling	Lab 3

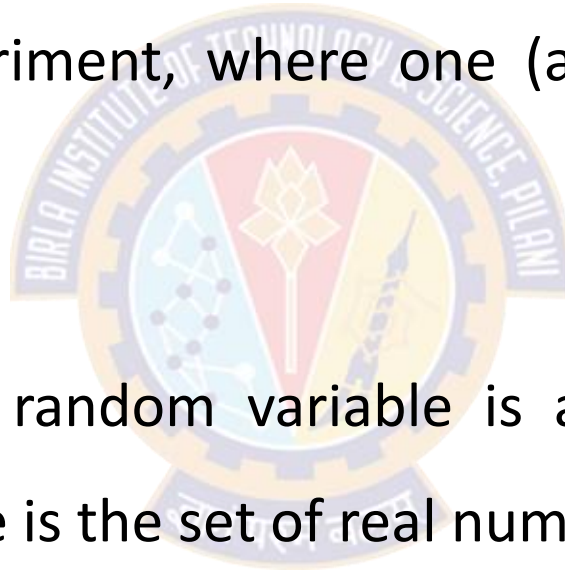
Agenda

- ☐ Random variables
- ☐ Discrete & continuous random variable,
- ☐ Probability Distribution Function
- ☐ Mean(Expectation) and variance of a random variable
- ☐ Joint distributions



Random Variables

- A **random variable** is a variable that assumes numerical values associated with the random outcome of an experiment, where one (and only one) numerical value is assigned to each sample point.
- In mathematical language, a random variable is a function whose domain is the sample space and whose range is the set of real numbers.



Types of Random Variables

- A **discrete random variable** can assume a countable number of values.
 - Number of steps to the top of the Eiffel Tower
 - Number of Heads obtain after tossing two coins simultaneously
- A **continuous random variable** can assume any value along a given interval of a number line.
 - The time a tourist stays at the top once s/he gets there.
 - The height of a person can take any value within a certain range (e.g., 150 cm to 200 cm),

Examples of Random Variables

➤ Discrete random variables

- Number of sales
- Number of calls
- Number of Shares of a stock
- Number of People in a line
- Number of Mistakes per page

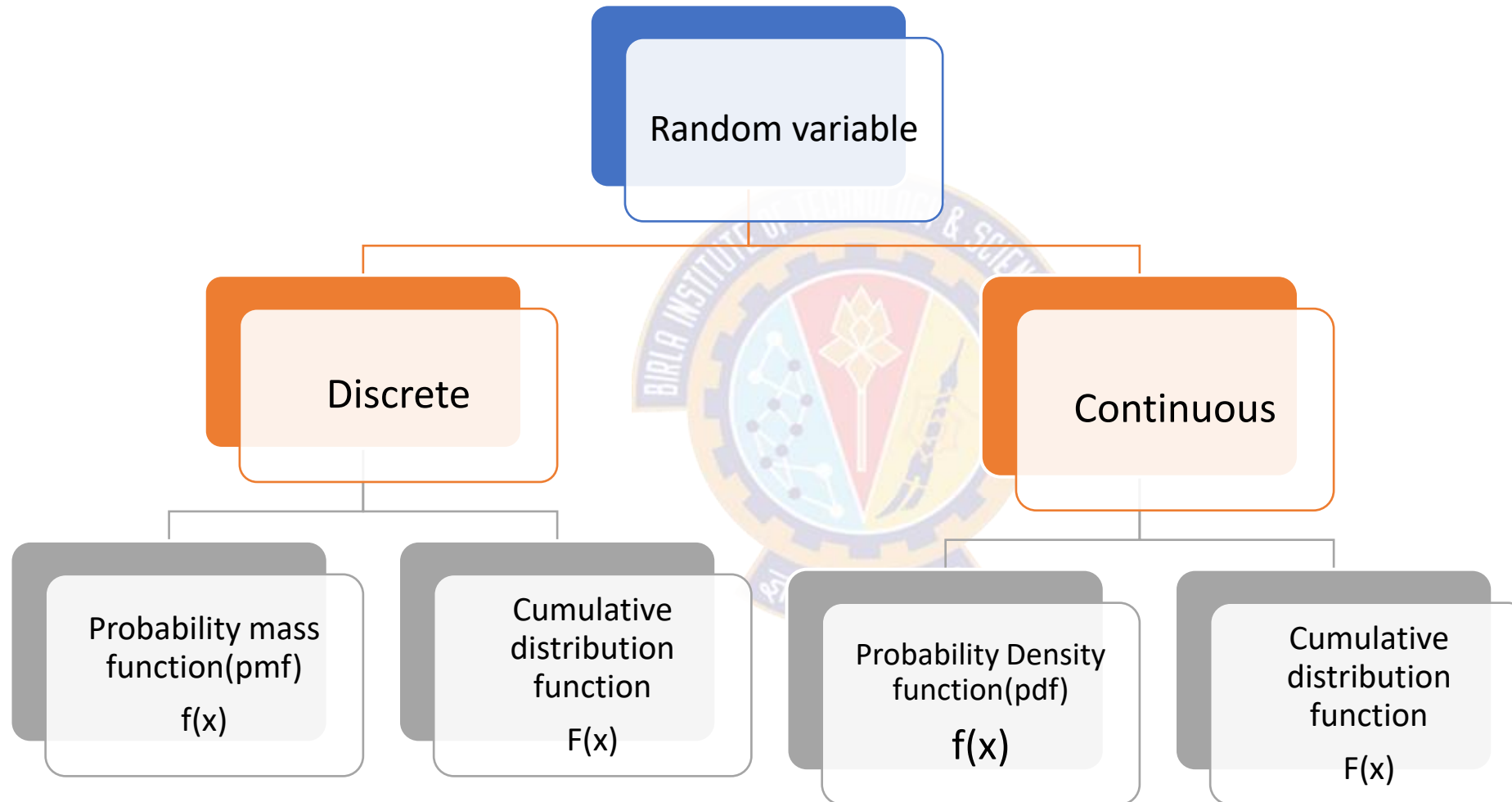


➤ Continuous random variables

- Length of each iron bar
- Depth of each bore well
- Volume of each cylinder
- Time to complete each project
- Weight of each newly born baby



Classification of Random Variables



➤ **Identify which of the following variables are discrete and which are continuous:**

- The number of books in a library. (Discrete / Continuous)
- The weight of a watermelon in kilograms. (Discrete / Continuous)
- The amount of rainfall in a day (in millimeters). (Discrete / Continuous)
- The number of passengers on a bus. (Discrete / Continuous)
- The temperature of a cup of coffee (in °C). (Discrete / Continuous)
- The number of goals scored in a football match. (Discrete / Continuous)
- The length of a bridge (in meters). (Discrete / Continuous)
- The number of steps you take in a day. (Discrete / Continuous)
- The time it takes to bake a cake (in minutes). (Discrete / Continuous)

Discrete Probability Distributions

- The probability distribution for a random variable describes how probabilities are distributed over the values of the random variable.
- The probability distribution is defined by a probability function, denoted by $f(x)$ or $p(x)$, which provides the probability for each value of the random variable.
- The required conditions for a discrete probability function are:
$$f(x) \geq 0$$
$$\sum f(x) = 1$$
- We can describe a discrete probability distribution with a **table, graph, or equation**.
- Advantage: once the probability distribution is known, it is relatively easy to determine the probability of a variety of events that may be of interest to the decision maker.

Example:

- Discrete random variable with a finite number of values
 - Let x = number of TV sets sold at the store in one day where x can take on 5 values (0, 1, 2, 3, 4)
- Discrete random variable with an infinite sequence of values
 - Let x = number of customers arriving in one day where x can take on the values 0, 1, 2, ...
 - We can count the customers arriving, but there is no finite upper limit on the number that might arrive.

Probability Distributions for Discrete Random Variables

- A company monitors employees' daily health status. Each day, an employee falls into one of the following mutually exclusive categories:

$X = 0 \rightarrow$ Healthy & Present (H)

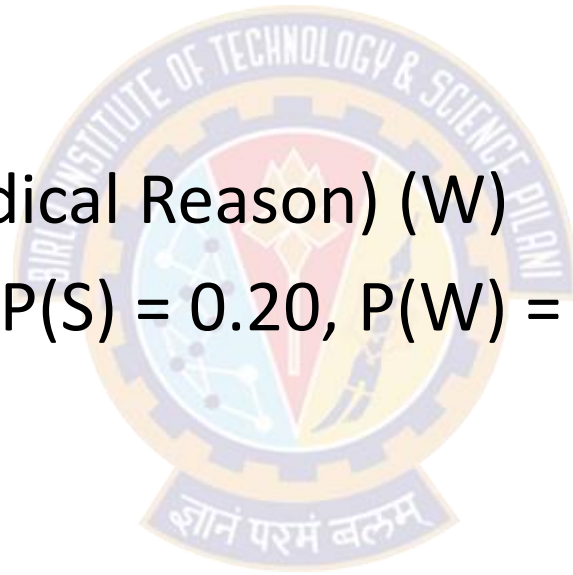
$X = 1 \rightarrow$ Sick Leave (S)

$X = 2 \rightarrow$ Work-from-Home (Medical Reason) (W)

Probabilities: $P(H) = 0.70, P(S) = 0.20, P(W) = 0.10$

Properties:

- $P(X) \geq 0$
- Sum of probabilities = 1



Discrete Uniform Probability Distribution

➤ The discrete uniform **probability** distribution is the simplest example of a discrete probability distribution given by a formula.

○ The discrete uniform probability function is

$$f(x) = 1/n$$

where:

n = the number of values the random variable may assume

- Note that the values of the random variable are equally likely.
- Example: A discrete RV X : No. on a dice has the uniform probability

$$f(x) = 1/6 \text{ for } x = 1, 2, 3, 4, 5, 6$$

Example

- Suppose a student takes **1, 2, or 3 attempts** to pass a driving test. Let the probability function be:

$$f(x) = \frac{4 - x}{6}, \quad x = 1, 2, 3$$

- Is this a valid probability function?



Mathematical Expectation

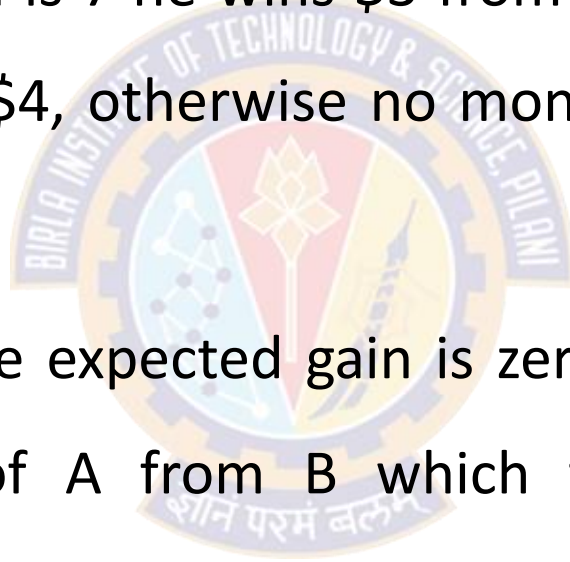
- Mean is a measure of location or central tendency in the sense that it roughly locates a middle or average value of the random variable
- The mean or Expected value of a discrete random variable:

$$E(X) = \mu = \sum x f(x)$$

Example:

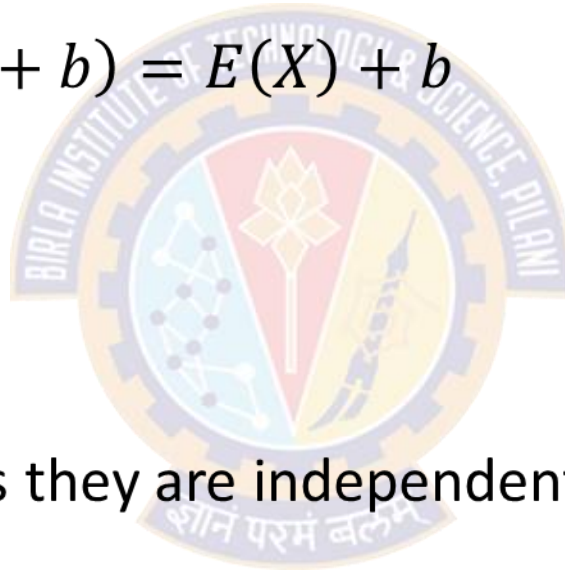
A game is played by two person A and B as follows:

- “A throws two dice. If the sum is 7 he wins \$3 from B. If the sum is 8, he loses \$2 and if the sum is 3, he loses \$4, otherwise no money changes hand. Is this a fair game?”
- A game is said to be fair if the expected gain is zero. In order to check it, let the random variable X = gain of A from B which takes values with probability distribution as follows:



Mathematical Expectation

- Multiplying RV by a constant a , $E(aX) = aE(X)$
- Adding a constant b , $E(X + b) = E(X) + b$
- Therefore, $E(aX + b) = ?$
- $E(X/Y) \neq \frac{E(X)}{E(Y)}$
- $E(X.Y) \neq E(X).E(Y)$ unless they are independent



Variability of Discrete Random Variables

➤ The **variance** of a **discrete random variable** x is

$$\text{var}(X) = \sigma^2 = E(X - \mu)^2 = \sum (x - \mu)^2 f(x)$$

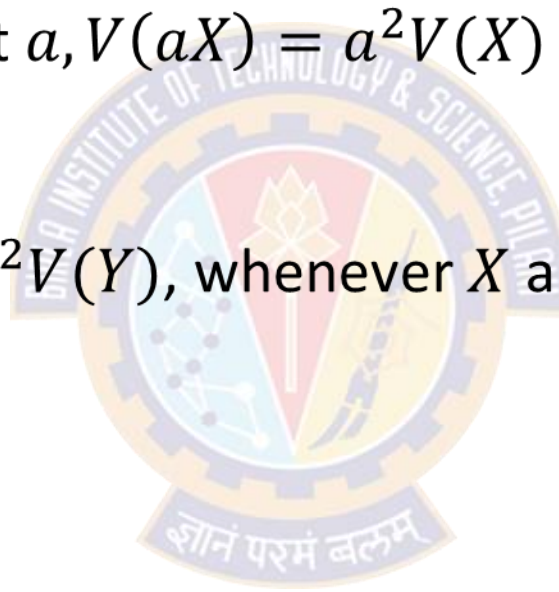
➤ Or

$$\begin{aligned}\text{var}(X) = \sigma^2 &= E(X^2) - (E(X))^2 \\ &= E(X^2) - \mu^2 \\ &= \sum x^2 f(x) - \mu^2\end{aligned}$$

➤ The **standard deviation** (σ), is defined as the positive square root of the variance.

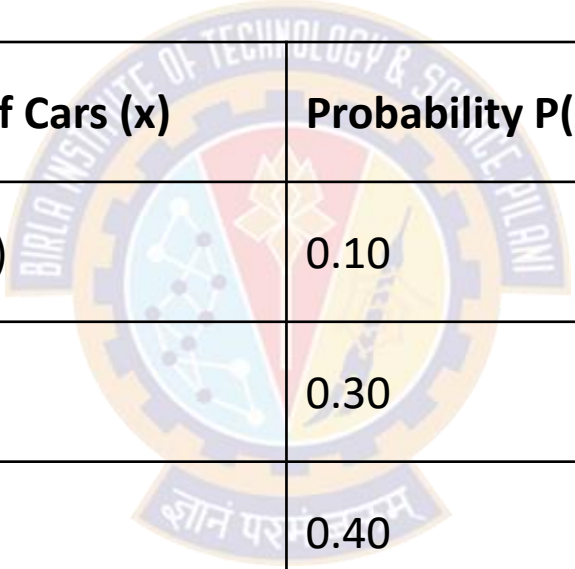
Rules of variability

- $V(b) = 0$, b is the constant
- Multiplying RV by a constant a , $V(aX) = a^2V(X)$
- Therefore, $V(aX+b) = ?$
- $V(aX + bY) = a^2V(X) + b^2V(Y)$, whenever X and Y are independent
- $\sigma_{aX} = |a|\sigma_x$



Example:

A fast-food restaurant manager analyzes how many cars are typically in line at the drive-thru at lunch hour. Let the **random variable X** represent the number of cars in line when a customer arrives. Find the mean, standard deviation and variance. Historical data shows:



Number of Cars (x)	Probability $P(X=x)$
0 (no wait)	0.10
1 car	0.30
2 cars	0.40
3 cars	0.15
4 cars	0.05

Expected value & SD:

x	$P(X = x)$	$x \cdot P(X)$	$(x - 1.75)^2 \cdot P(X)$
0	0.10	0.00	$(0 - 1.75)^2 \cdot 0.10 = 3.0625 \cdot 0.10 = 0.30625$
1	0.30	0.30	$(1 - 1.75)^2 \cdot 0.30 = 0.5625 \cdot 0.30 = 0.16875$
2	0.40	0.80	$(2 - 1.75)^2 \cdot 0.40 = 0.0625 \cdot 0.40 = 0.025$
3	0.15	0.45	$(3 - 1.75)^2 \cdot 0.15 = 1.5625 \cdot 0.15 = 0.234375$
4	0.05	0.20	$(4 - 1.75)^2 \cdot 0.05 = 5.0625 \cdot 0.05 = 0.253125$
Total	1.00	1.75	0.9875 (Variance σ^2)

Standard Deviation $\downarrow \sqrt{0.9875} \approx \boxed{0.9937}$

Cumulative Probability distribution Function

If X is a discrete random variable which assumes values $x = x_1, x_2, x_3, \dots, x_i, \dots, x_n$ with their corresponding probabilities $f(x) = f_1, f_2, f_3, \dots, f_i, \dots, f_n$ respectively then the cumulative distribution function (CDF) of the RV X is defined as $F(x) = P(X \leq x)$.

Specifically,

$$F(x_1) = P(X \leq x_1) = P(X=x_1) = f_1$$

$$F(x_2) = P(X \leq x_2) = P(X=x_1) + P(X=x_2) = f_1 + f_2$$

$$F(x_3) = P(X \leq x_3) = P(X=x_1) + P(X=x_2) + P(X=x_3) = f_1 + f_2 + f_3$$

.....

.....

$$F(x_n) = P(X \leq x_n) = P(X=x_1) + P(X=x_2) + \dots + P(X=x_n) = f_1 + f_2 + \dots + f_n = \underset{23}{1} \text{ (=Total probability)}$$

Cumulative Distribution for Discrete Data

Cumulative probability Distribution:

It is given by $F(x) = P(X \leq x)$.

Example: If two dices are rolled. Find the probability distribution and cumulative probability distribution. Also represent in graph.

Solution: X- random variable is sum of the two numbers

X	1	2	3	4	5	6	7	8	9	10	11	12
P(x)	-	$\frac{1}{36}$	$\frac{2}{36}$	$\frac{3}{36}$	$\frac{4}{36}$	$\frac{5}{36}$	$\frac{6}{36}$	$\frac{5}{36}$	$\frac{4}{36}$	$\frac{3}{36}$	$\frac{2}{36}$	$\frac{1}{36}$
F(x)	0	$\frac{1}{36}$	$\frac{3}{36}$	$\frac{6}{36}$	$\frac{10}{36}$	$\frac{15}{36}$	$\frac{21}{36}$	$\frac{26}{36}$	$\frac{30}{36}$	$\frac{33}{36}$	$\frac{35}{36}$	1

Example: 1

A hospital has **4 patients**:

A – Diabetic

B – Non-diabetic

C – Diabetic

D – Non-diabetic

So there are **2 diabetic** and **2 non-diabetic** patients.

The hospital randomly selects **2 patients** for a screening study.

Let the random variable:

X = number of diabetic patients in the selected group of 2

$S = \{AB, AC, AD, BC, BD, CD\}$

$$P(X = 0) = \frac{1}{6}, \quad P(X = 1) = \frac{4}{6}, \quad P(X = 2) = \frac{1}{6}$$

Solution:

Expected Value (Mean)

$$E(X) = \sum xP(x)$$

$$E(X) = 0 \cdot \frac{1}{6} + 1 \cdot \frac{4}{6} + 2 \cdot \frac{1}{6}$$

$$E(X) = \frac{6}{6} = 1$$

Variance

$$\text{Var}(X) = E(X^2) - [E(X)]^2$$

$$E(X^2) = 0^2 \frac{1}{6} + 1^2 \frac{4}{6} + 2^2 \frac{1}{6}$$

$$E(X^2) = 0 + \frac{4}{6} + \frac{4}{6} = \frac{8}{6} = \frac{4}{3}$$

$$\text{Var}(X) = \frac{4}{3} - 1^2 = \frac{4}{3} - 1 = \frac{1}{3}$$

Example: 2

A random variable X has the following probability function:

X	0	1	2	3	4
$P(X = x)$	k	$2k$	$3k$	$2k$	k

Find

- i) the value of k
- ii) $P(X < 3)$
- iii) $P(0 < X < 4)$
- iv) the distribution function of X
- v) $E(X)$, vi) $Var(X)$



Solution:

i) Find the value of k

Since the total probability must equal 1:

$$k + 2k + 3k + 2k + k = 9k = 1 \Rightarrow k = \frac{1}{9}$$

ii) Find $P(X < 3)$

$$P(X < 3) = P(X = 0) + P(X = 1) + P(X = 2) = k + 2k + 3k = 6k = 6 \cdot \frac{1}{9} = \frac{2}{3}$$



iii) Find $P(0 < X < 4)$

$$P(0 < X < 4) = P(X = 1) + P(X = 2) + P(X = 3) = 2k + 3k + 2k = 7k = 7 \cdot \frac{1}{9} = \frac{7}{9}$$

iv) Find the **distribution function (CDF)** of X

Let $F(x) = P(X \leq x)$

x	$F(x)$
$x < 0$	0
0	$k = \frac{1}{9}$
1	$k + 2k = \frac{3}{9}$
3	$k + 2k + 3k + 2k = \frac{8}{9}$
4 or more	1

v) Find the **Expected Value** $E(X)$

$$\begin{aligned} E(X) &= \sum x \cdot P(X = x) = 0 \cdot k + 1 \cdot 2k + 2 \cdot 3k + 3 \cdot 2k + 4 \cdot k \\ &= 0 + 2k + 6k + 6k + 4k = 18k = 18 \cdot \frac{1}{9} = 2 \end{aligned}$$



vi) Find the **Variance** $\text{Var}(X)$

First, compute $E(X^2)$:

$$\begin{aligned} E(X^2) &= \sum x^2 \cdot P(X = x) = 0^2 \cdot k + 1^2 \cdot 2k + 2^2 \cdot 3k + 3^2 \cdot 2k + 4^2 \cdot k \\ &= 0 + 2k + 12k + 18k + 16k = 48k = 48 \cdot \frac{1}{9} = \frac{16}{3} \end{aligned}$$

$$\text{Var}(X) = E(X^2) - [E(X)]^2 = \frac{16}{3} - 2^2 = \frac{16}{3} - 4 = \frac{4}{3}$$

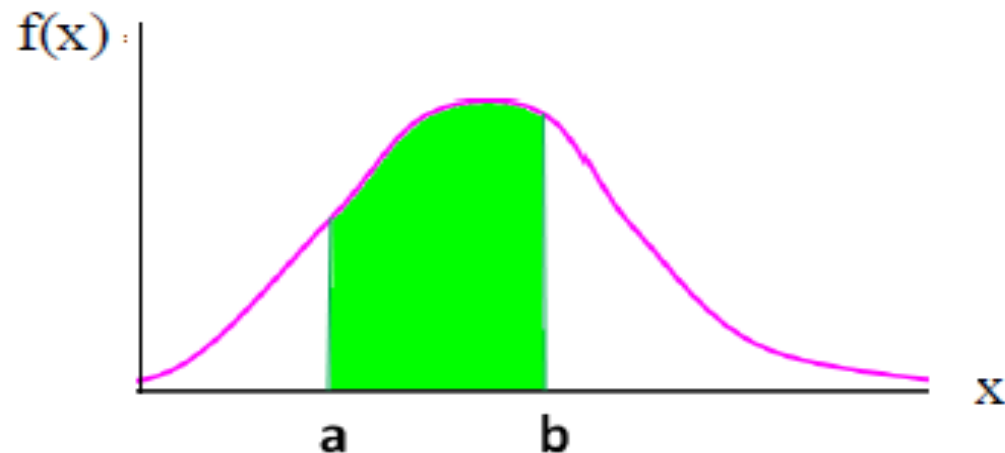
Continuous Probability Distributions

- A continuous random variable can assume any value in an interval on the real line or in a collection of intervals.
- It is not possible to talk about the probability of the random variable assuming a particular value.
- Instead, we talk about the probability of the random variable assuming a value within a given interval.
- The probability of the random variable assuming a value within some given interval from x_1 to x_2 is defined to be the area under the graph of the probability density function between x_1 and x_2 .

Continuous Random Variables

A continuous random variable can assume any value in an interval on the real line or in a collection of intervals.

The probability of the random variable assuming a value within some given interval from x_1 to x_2 is defined to be the area under the graph of the **probability density function** between x_1 and x_2



Example:

- Height of students in a class
- Amount of ice tea in a glass
- Change in temperature throughout a day
- Price of a car in next year

Continuous Random Variables

Probability Density Function

For a continuous random variable X , a **probability density function** is a function such that

$$(1) \quad f(x) \geq 0$$

$$(2) \quad \int_{-\infty}^{\infty} f(x) dx = 1$$

$$(3) \quad P(a \leq X \leq b) = \int_a^b f(x) dx = \text{area under } f(x) \text{ from } a \text{ to } b \text{ for any } a \text{ and } b \quad (4.1)$$

Continuous Random Variables

Mean and Variance

Suppose that X is a continuous random variable with probability density function $f(x)$. The **mean** or **expected value** of X , denoted as μ or $E(X)$, is

$$\mu = E(X) = \int_{-\infty}^{\infty} xf(x)dx$$

The **variance** of X , denoted as $V(X)$ or σ^2 , is

$$\sigma^2 = V(X) = \int_{-\infty}^{\infty} (x - \mu)^2 f(x)dx = \int_{-\infty}^{\infty} x^2 f(x)dx - \mu^2$$

The **standard deviation** of X is $\sigma = \sqrt{\sigma^2}$.

Continuous Random Variables

EXAMPLE 1

Calculating probabilities from the probability density function

If a random variable has the probability density

$$f(x) = \begin{cases} 2e^{-2x} & \text{for } x > 0 \\ 0 & \text{for } x \leq 0 \end{cases}$$

find the probabilities that it will take on a value

- (a) between 1 and 3;
- (b) greater than 0.5.



Solution Evaluating the necessary integrals, we get

(a)
$$\int_1^3 2e^{-2x} dx = e^{-2} - e^{-6} = 0.133$$

(b)
$$\int_{0.5}^{\infty} 2e^{-2x} dx = e^{-1} = 0.368$$

With reference to the preceding example, find the distribution function and use it to determine the probability that the random variable will take on a value less than or equal to 1.

Performing the necessary integrations, we get

$$F(x) = \begin{cases} 0 & \text{for } x \leq 0 \\ \int_0^x 2e^{-2t} dt = 1 - e^{-2x} & \text{for } x > 0 \end{cases}$$

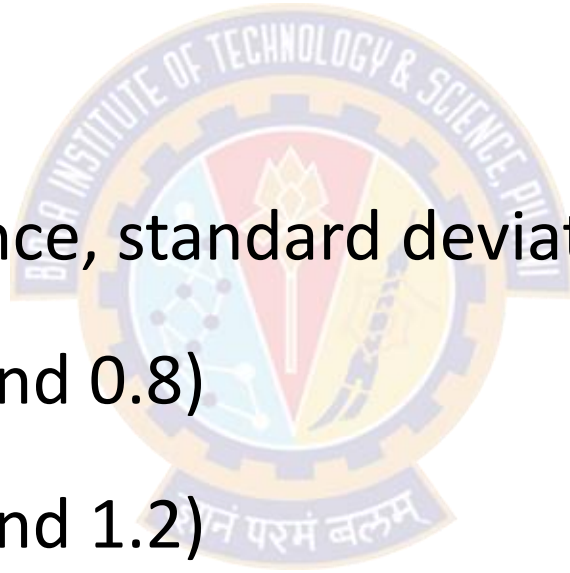
and substitution of $x = 1$ yields

$$F(1) = 1 - e^{-2} = 0.865$$



Example 2

- The Pdf of a random variable is given by $f(x) = x$ for $0 < x < 1$
 $= 2-x$ for $1 \leq x < 2$
 $= 0$, elsewhere
- Determine, mean, variance, standard deviation
 - $P(x \text{ lies in between } 0.2 \text{ and } 0.8)$
 - $P(x \text{ lies in between } 0.6 \text{ and } 1.2)$



Solution

- The Pdf of a random variable is given by $f(x) = x$ for $0 < x < 1$
 $= 2 - x$ for $1 \leq x < 2$
 $= 0$, elsewhere
- Determine, mean, variance, standard deviation
 - $P(x \text{ lies in between } 0.2 \text{ and } 0.8)$
 - $P(x \text{ lies in between } 0.6 \text{ and } 1.2)$

Solution

$$\begin{aligned}\text{Mean} = \mu = E(X) &= \int_{-\infty}^{\infty} x f(x) dx = \int_0^1 x x dx + \int_1^2 x(2 - x) dx \\ &= \int_0^1 x^2 dx + 2 \int_1^2 x dx - \int_1^2 x^2 dx = \left. \frac{x^3}{3} \right|_0^1 + 2 \left. \frac{x^2}{2} \right|_1^2 - \left. \frac{x^3}{3} \right|_1^2 = \frac{1}{3} + 3 - \frac{7}{3} = 1\end{aligned}$$

$$\begin{aligned}E(X^2) &= \int_{-\infty}^{\infty} x^2 f(x) dx = \int_0^1 x^2 x dx + \int_1^2 x^2(2 - x) dx = \int_0^1 x^3 dx + 2 \int_1^2 x^2 dx - \int_1^2 x^3 dx \\ &= \left. \frac{x^4}{4} \right|_0^1 + 2 \left. \frac{x^3}{3} \right|_1^2 - \left. \frac{x^4}{4} \right|_1^2 \\ &= \frac{1}{4} + \frac{14}{3} - \frac{15}{4} = \frac{7}{6}\end{aligned}$$

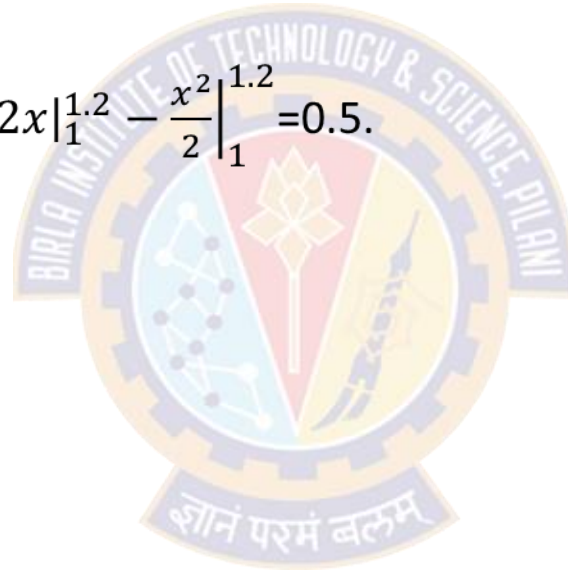
$$\text{Variance} = \sigma^2 = \int_{-\infty}^{\infty} x^2 f(x) dx - \mu^2 = \frac{7}{6} - (1)^2 = \frac{1}{6}$$

$$\text{SD} = \sigma = +\sqrt{\text{variance}} = +\sqrt{\frac{1}{6}} = \frac{1}{\sqrt{6}} = 0.4082$$

➤ $P(x \text{ lies in between } 0.2 \text{ and } 0.8) = P(0.2 < X < 0.8) = \int_{0.2}^{0.8} f(x) dx = \int_{0.2}^{0.8} x dx = \left. \frac{x^2}{2} \right|_{0.2}^{0.8} = 0.3$

➤ $P(x \text{ lies in between } 0.6 \text{ and } 1.2) = P(0.6 < X < 1.2) = P(0.6 < X < 1) + P(1 < X < 1.2) =$

$$= \int_{0.6}^1 x dx + \int_1^{1.2} (2 - x) dx = \left. \frac{x^2}{2} \right|_{0.6}^1 + 2x \Big|_1^{1.2} - \left. \frac{x^2}{2} \right|_1^{1.2} = 0.5.$$



Example 3:

A probability density function assigns probability one to $(-\infty, \infty)$

Find k so that the following can serve as the probability density of a random variable:

$$f(x) = \begin{cases} 0 & \text{for } x \leq 0 \\ kxe^{-4x^2} & \text{for } x > 0 \end{cases}$$



Solution

$$\int_{-\infty}^{\infty} f(x) dx = \int_0^{\infty} kxe^{-4x^2} dx = \int_0^{\infty} \frac{k}{8} \cdot e^{-u} du = \frac{k}{8} = 1$$

so that $k = 8$.

Practice Problems

Problem-1

A function is defined as

$$f(x) = kx^2, \quad 0 \leq x \leq 2$$

and $f(x) = 0$ otherwise.

- (a) Find the value of k so that $f(x)$ is a valid probability density function.
- (b) Find $P(1 \leq x \leq 2)$.

Solution

(a) Condition for a valid PDF:

$$\int_0^2 kx^2 dx = 1$$

$$k \left[\frac{x^3}{3} \right]_0^2 = 1$$

$$k \cdot \frac{8}{3} = 1 \quad \Rightarrow \quad k = \frac{3}{8}$$

(b) Find probability:

$$P(1 \leq x \leq 2) = \int_1^2 \frac{3}{8} x^2 dx$$

$$= \frac{3}{8} \left[\frac{x^3}{3} \right]_1^2 = \frac{1}{8} (8 - 1) = \frac{7}{8}$$

Problem 2:

The length of satisfactory service (years) provided by a certain model of laptop computer is random variable having the probability density function :

$$f(x) = \begin{cases} \frac{1}{4.5} e^{-\frac{x}{4.5}} & x > 0 \\ 0 & \text{otherwise} \end{cases}$$

Find the probabilities that one of these laptops will provide satisfactory service for

- a) At most 2.5 years (Ans: 0.4262)
- b) Anywhere from 4 to 6 years (Ans: 0.1475)
- c) At least 6.75 years (Ans: 0.2231)

Problem-3

Let a continuous random variable X have PDF

$$f(x) = \lambda e^{-\lambda x}, \quad x \geq 0$$

where $\lambda = 2$.

Find:

(a) $P(X \leq 1)$

(b) $P(1 < X < 3)$

Solution: (a) $1 - e^{-3}$
 (b) $e^{-3} - e^{-6}$

Problem:4

For following probability density function :

$$f(x) = \begin{cases} x & 0 \leq x \leq 1 \\ 2 - x & 1 \leq x < 2 \\ 0 & \text{otherwise} \end{cases} . \text{ Find the mean and variance.}$$

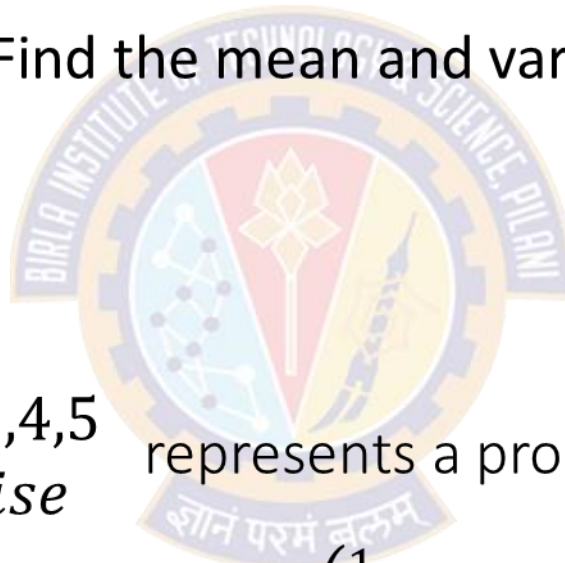
Ans: Mean =1, Variance = $\frac{1}{6}$

Problem:5

If $P(X = x) = \begin{cases} kx & x = 1,2,3,4,5 \\ 0 & \text{otherwise} \end{cases}$ represents a probability function , find i) k ,

ii) $P(X \text{ being a prime number})$, iii) $P\left(\frac{1}{2} < X < \frac{5}{2}\right)$.

(Answer: i) $k = \frac{1}{15}$, ii) $\frac{11}{15}$ iii) $\frac{1}{5}$)



Problem:6

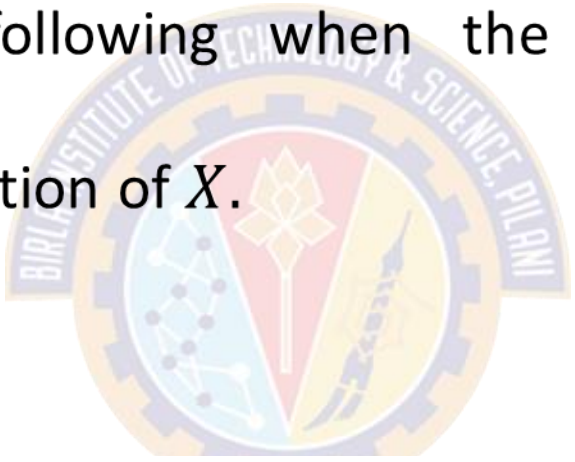
From a lot of 10 items containing 3 defective items, a sample of 4 items is drawn at random. Let the random variable X denote the number of defective items in the sample. Answer the following when the sample is drawn without replacement:

a) Find the probability distribution of X .

b) Find $P(X \leq 1)$

c) Find $P(0 < X < 2)$

Ans: a)



x	0	1	2	3
$P(X)$	$\frac{1}{6}$	$\frac{1}{2}$	$\frac{3}{10}$	$\frac{1}{30}$

b) $\frac{2}{3}$ c) $\frac{1}{2}$

Joint Probability Distribution Function

- ❖ Let $X = \{x_1, x_2, \dots, x_m\}$ and $Y = \{y_1, y_2, \dots, y_n\}$ be two discrete random variables. Then $P(x, y) = J_{ij}$ is called joint probability function of X and Y if it satisfies the conditions:

$$(i) J_{ij} \geq 0 \quad (ii) \sum_{i=1}^m \sum_{j=1}^n J_{ij} = 1$$

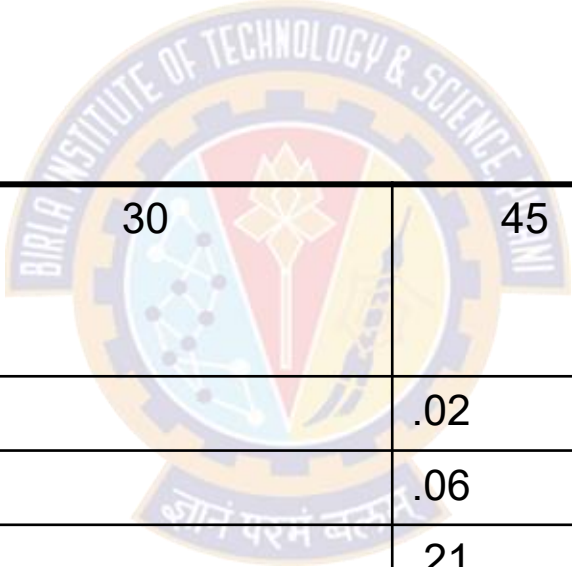
- ❖ Set of values of this joint probability function J_{ij} is called joint probability distribution of X and Y.

	y_1	y_2	...	y_n	Sum
x_1	J_{11}	J_{12}	...	J_{1n}	$f(x_1)$
x_2	J_{21}	J_{22}	...	J_{2n}	$f(x_2)$
...	...	...	...	...	...
x_m	J_{m1}	J_{m2}	...	J_{mn}	$f(x_m)$
Sum	$g(y_1)$	$g(y_2)$	...	$g(y_n)$	Total = 1

- If X and Y are discrete random variables, the joint probability distribution of X and Y is a description of the set of points (x,y) in the range of (X,Y) along with the probability of each point.
- The joint probability distribution of two discrete random variables is sometimes referred to as the **bivariate probability distribution** or **bivariate distribution**.
- Thus, we can describe the joint probability distribution of two discrete random variables is through a **joint probability mass function**

$$f(x,y)=P(X=x,Y=y)$$

➤ We often want to determine the joint probability of two variables, such as **X and Y**. Suppose we are able to determine the following information for education (X) and age (Y) for all Indian citizens based on the census.



Age (Y):		30	45	70
Education (X)				
None	0	.01	.02	.05
Primary	1	.03	.06	.10
Secondary	2	.18	.21	.15
College	3	.07	.08	.04

➤ Each cell is the relative frequency (f/N).

➤ We can define the joint probability distribution as: $p(x, y) = \Pr(X = x \text{ and } Y = y)$

Example: what is the probability of getting a 30 year old college graduate?

$$p(x, y) = \Pr(X=3 \text{ and } Y=30) = .07$$

We can see that: $p(x) = \sum_y p(x, y)$

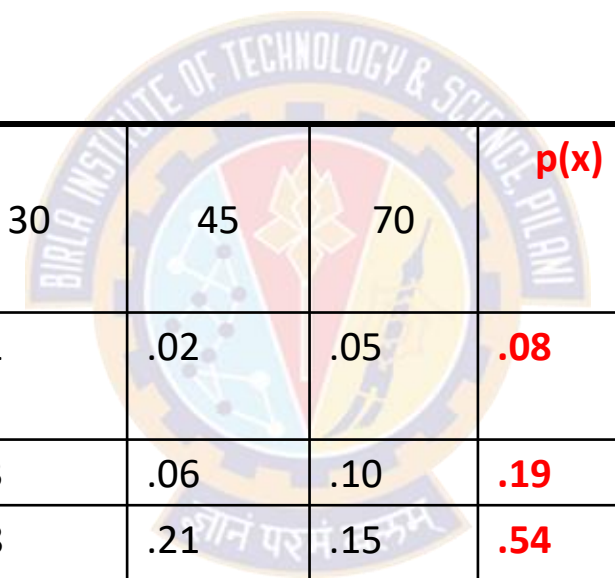
$$p(x=1) = .03 + .06 + .10 = .19$$



Age (Y):		30	45	70
Education (X)				
None	0	.01	.02	.05
Primary	1	.03	.06	.10
Secondary	2	.18	.21	.15
College	3	.07	.08	.04

Marginal Probability

- We call this the **marginal probability** because it is calculated by summing across rows or columns and is thus reported in the margins of the table.
- We can calculate this for our entire table.



Age (Y): Education (X)	30	45	70	$p(x)$
None: 0	.01	.02	.05	.08
Primary: 1	.03	.06	.10	.19
Secondary: 2	.18	.21	.15	.54
College: 3	.07	.08	.04	.19
$p(y)$	.29	.37	.34	1

Joint Density Function

When X and Y are continuous random variables, the **joint density function** $f(x, y)$ is a surface lying above the xy plane, and $P[(X, Y) \in A]$, where A is any region in the xy plane, is equal to the volume of the right cylinder bounded by the base A and the surface.



The function $f(x, y)$ is a **joint density function** of the continuous random variables X and Y if

1. $f(x, y) \geq 0$, for all (x, y) ,
2. $\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x, y) \, dx \, dy = 1$,
3. $P[(X, Y) \in A] = \int \int_A f(x, y) \, dx \, dy$, for any region A in the xy plane.

Marginal Distributions

The **marginal distributions** of X alone and of Y alone are

$$g(x) = \sum_y f(x, y) \quad \text{and} \quad h(y) = \sum_x f(x, y)$$

for the discrete case, and

$$g(x) = \int_{-\infty}^{\infty} f(x, y) \, dy \quad \text{and} \quad h(y) = \int_{-\infty}^{\infty} f(x, y) \, dx$$

for the continuous case.



Independent Random Variables

- Let X and Y are two random variables with joint probability function $f(x, y)$ are said to be independent if following condition satisfied:
- $f(x, y) = g(x).h(y)$ for all x and y , where $g(x)$ is marginal probability function of X and $h(y)$ is marginal probability function of Y .

Example: Suppose $f(x, y) = e^{-(x+y)}$, $x \geq 0, y \geq 0$. For this probability function, marginal probability function for X is $g(x) = e^{-x}$, $x \geq 0$ and marginal probability function for Y is $h(y) = e^{-y}$, $y \geq 0$.

Clearly $f(X, Y) = g(X).h(Y)$. So X and Y are independent variables.

Conditional Probability Distribution Function

- **Discrete Case:**

$$P[X = x_i / Y = y_j] = \frac{P[X = x_i / Y = y_j]}{P[Y = y_j]} = \frac{p_{ij}}{p_{.j}}$$

The set $\{x_i, p_{ij} / p_{.j}\}$, $i = 1, 2, 3, \dots$ is called the conditional probability distribution of X given $Y = y_j$.

The conditional probability function of Y given $X = x_i$ is given by

$$P[Y = y_i / X = x_j] = \frac{P[Y = y_i / X = x_j]}{P[X = x_j]} = \frac{p_{ij}}{p_{i.}}$$

The set $\{y_i, p_{ij} / p_{i.}\}$, $j = 1, 2, 3, \dots$ is called the conditional probability distribution of Y given $X = x_i$.

Conditional Probability Density Function

$$f(y | x) = \frac{f(x, y)}{f_X(x)} \quad \text{or} \quad f(x | y) = \frac{f(x, y)}{f_Y(y)}$$

- **the continuous version of Bayes' theorem**

$$f(y | x) = \frac{f(x | y)f_Y(y)}{f_X(x)}$$

- **another expression of the marginal pdf**

$$f_X(x) = \int_{-\infty}^{\infty} f(x, y)dy = \int_{-\infty}^{\infty} f(x | y)f_Y(y)dy$$

$$f_Y(y) = \int_{-\infty}^{\infty} f(x, y)dx = \int_{-\infty}^{\infty} f(y | x)f_X(x)dx$$

Example 1:

You want to compute the probability that a patient has the **flu** given they have a **fever**.

Let:

- $D \in \{\text{Flu}, \text{No Flu}\}$: Disease (target variable)
- $S \in \{\text{Fever}, \text{No Fever}\}$: Symptom (observed variable)

You are given the **joint distribution** from past data:

Disease	Symptom	$P(D = d, S = s)$
Flu	Fever	0.30
Flu	No Fever	0.10
No Flu	Fever	0.10
No Flu	No Fever	0.50

Compute the **conditional probability distribution**: $P(D|S=\text{Fever})$

Solution

- Compute the marginal probability $P(S=\text{Fever})$:

$$P(S = \text{Fever}) = P(\text{Flu}, \text{Fever}) + P(\text{No Flu}, \text{Fever}) = 0.30 + 0.10 = 0.40$$

- Compute the Conditional probability:

$$P(D = d \mid S = \text{Fever}) = \frac{P(D = d, S = \text{Fever})}{P(S = \text{Fever})}$$

Final Answer:

- $P(\text{Flu} \mid \text{Fever}) = \frac{0.30}{0.40} = 0.75$
- $P(\text{No Flu} \mid \text{Fever}) = \frac{0.10}{0.40} = 0.25$

Example 2:

Consider the joint distribution of X and Y .

Compute the following probabilities:

(i) $P(X = 1, Y = 2)$ (ii) $P(X \geq 1, Y \geq 2)$

(iii) $P(X \leq 1, Y \leq 2)$ (iv) $P(X + Y \geq 2)$ (v) $P(X \geq 1, Y \leq 2)$.

$X \backslash Y$	0	1	2	3
0	0	$\frac{1}{8}$	$\frac{1}{4}$	$\frac{1}{8}$
1	$\frac{1}{8}$	$\frac{1}{4}$	$\frac{1}{8}$	0

Solution:

(i) $X = \{0, 1\}, Y = \{0, 1, 2, 3, 4\}$

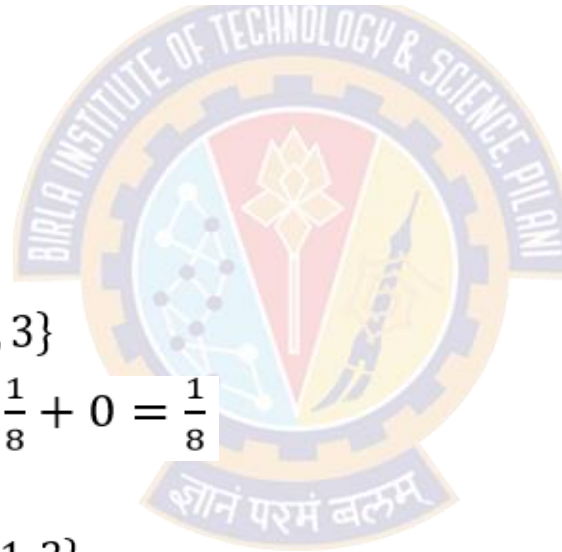
$$P(X = 1, Y = 2) = P(1, 2) = \frac{1}{8}$$

(ii) If $X \geq 1, X = \{1\}$. If $Y \geq 2, Y = \{2, 3\}$

$$P(X \geq 1, Y \geq 2) = P(1, 2) + P(1, 3) = \frac{1}{8} + 0 = \frac{1}{8}$$

(iii) If $X \leq 1, X = \{0, 1\}$. If $Y \leq 2, Y = \{0, 1, 2\}$

$$\begin{aligned} P(X \leq 1, Y \leq 2) &= P(0, 0) + P(0, 1) + P(0, 2) + P(1, 0) + P(1, 1) + P(1, 2) \\ &= 0 + \frac{1}{4} + \frac{1}{8} + \frac{1}{8} + \frac{1}{4} + \frac{1}{8} = \frac{7}{8} \end{aligned}$$



Cont.

(iv) If $X + Y \geq 2$ then

$$X + Y = 0 + 2 \text{ or } 0 + 3 \text{ or } 1 \text{ or } 1 + 2 \text{ or } 1 + 3$$

$$\begin{aligned} P(X + Y \geq 2) &= P(0, 2) + P(0, 3) + P(1, 1) + P(1, 2) + P(1, 3) \\ &= \frac{1}{4} + \frac{1}{8} + \frac{1}{4} + \frac{1}{8} + 0 = \frac{3}{4} \end{aligned}$$

(v) If $X \geq 1, X = \{1\}$. If $Y \leq 2, Y = \{0, 1, 2\}$

$$\begin{aligned} P(X \geq 1, Y \leq 2) &= P(1, 0) + P(1, 1) + P(1, 2) \\ &= \frac{1}{8} + \frac{1}{4} + \frac{1}{8} = \frac{1}{2} \end{aligned}$$

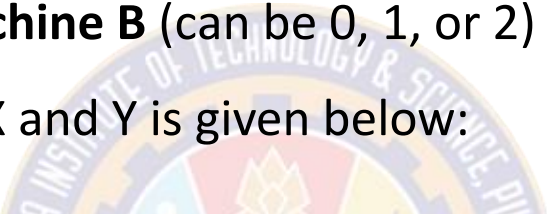
Example 3:

➤ Two machines in a factory produce defective items. Let:

X: Number of defective items from **Machine A** (can be 0 or 1)

Y: Number of defective items from **Machine B** (can be 0, 1, or 2)

The joint probability mass function of X and Y is given below:



$X \setminus Y$	0	1	2
0	0.10	0.20	0.10
1	0.10	0.30	0.20

Compute:

1. Find the **marginal distributions** of X and Y
2. Find $P(X = 1, Y = 2)$ Ans: 0.20
3. Find $P(Y = 1 \mid X = 0)$ Ans: 0.5
4. Find the **expected values** $E(X)$ and $E(Y)$ Ans: 0.6 and 1.10
5. Are X and Y independent?

Solution:

3. $P(Y = 1 \mid X = 0)$

$$P(Y = 1 \mid X = 0) = \frac{P(X = 0, Y = 1)}{P(X = 0)} = \frac{0.20}{0.40} = 0.5$$

5. Check: is $P(X = 0, Y = 1) = P(X = 0) \cdot P(Y = 1)$?

- $P(X = 0, Y = 1) = 0.20$
- $P(X = 0) = 0.40, P(Y = 1) = 0.50$
- $P(X = 0) \cdot P(Y = 1) = 0.40 \cdot 0.50 = 0.20$

✓ For this pair, they match. Try another:

- $P(X = 1, Y = 0) = 0.10$
- $P(X = 1) \cdot P(Y = 0) = 0.60 \cdot 0.20 = 0.12$

✗ They don't match. So X and Y are not independent.

Example 4:

- Find the joint distribution of X and Y which are the independent random variables with the following respective distributions.

x_i	1	2
$f(x_i)$	0.7	0.3

y_j	-2	5	8
$g(y_j)$	0.3	0.5	0.2

Solution:

Since X and Y are independent random variables,

$$J_{ij} = f(x_i)g(y_j)$$

$x \backslash y$	-2	5	8	$f(x)$
1	0.21	0.35	0.14	0.7
2	0.09	0.15	0.06	0.3
$g(y)$	0.3	0.5	0.2	Total = 1

Example 5:

A machine learning model is trained to classify images of **animals**. In the dataset:

- Each image is labeled based on whether the animal is **wild** or **domestic**
- Each image is also tagged for visual features like **fur**, **claws**, and **ears**

➤ Let:

X: Proportion of **wild animal images** in which **claws are detected**

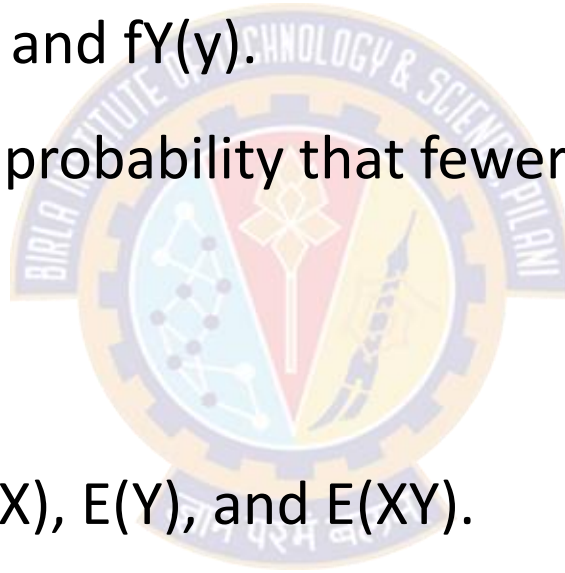
Y: Proportion of **domestic animal images** in which **claws are detected**

Suppose the **joint probability density function** for the proportions X and Y is given by:

Suppose the joint probability density function for the proportions X and Y is given by:

$$f(x, y) = \begin{cases} 8xy & \text{for } 0 \leq x \leq 1, 0 \leq y \leq 1 \\ 0 & \text{otherwise} \end{cases}$$

- ✓ Verify that $f(x, y)$ is a valid joint probability density function
- ✓ Find the marginal densities $f_X(x)$ and $f_Y(y)$.
- ✓ Compute $P(X < 0.5, Y > 0.5)$ — i.e., probability that fewer wild images but more domestic images show claws
- ✓ Are X and Y independent?
- ✓ Compute the expected values $E(X)$, $E(Y)$, and $E(XY)$.



Solution:

1. Verify $f(x, y)$ is a valid joint PDF

We need to check:

$$\iint_{\text{all } x, y} f(x, y) dx dy = 1$$

2. Find marginal densities

$$f_X(x) = \int_0^1 f(x, y) dy = \int_0^1 4xy dy = 4x \times \frac{y^2}{2} \Big|_0^1 = 4x \times \frac{1}{2} = 2x, \quad 0 \leq x \leq 1$$

$$f_Y(y) = \int_0^1 f(x, y) dx = \int_0^1 4xy dx = 4y \times \frac{x^2}{2} \Big|_0^1 = 4y \times \frac{1}{2} = 2y, \quad 0 \leq y \leq 1$$

Solution:

3. **Compute** $P(X < 0.5, Y > 0.5)$

$$P = \int_0^{0.5} \int_{0.5}^1 4xy \, dy \, dx = 0.1875$$

4. **Are X and Y independent?**

Check if

$$f(x, y) \stackrel{?}{=} f_X(x) \cdot f_Y(y)$$

Compute:

$$f_X(x) \cdot f_Y(y) = (2x)(2y) = 4xy$$

Which equals $f(x, y)$.

Thus, X and Y are independent.

Solution:

5. Calculate $E(X)$, $E(Y)$, and $E(XY)$

- $E(X)$:

$$E(X) = \int_0^1 x f_X(x) dx = \int_0^1 x \cdot 2x dx = 2 \int_0^1 x^2 dx = 2 \times \frac{1}{3} = \frac{2}{3} \approx 0.6667$$

- $E(Y)$:

$$E(Y) = \int_0^1 y f_Y(y) dy = \int_0^1 y \cdot 2y dy = 2 \int_0^1 y^2 dy = 2 \times \frac{1}{3} = \frac{2}{3} \approx 0.6667$$


Since independent,

$$E(XY) = E(X)E(Y) = \left(\frac{2}{3}\right) \times \left(\frac{2}{3}\right) = \frac{4}{9} \approx 0.4444$$

Example 6:

The joint probability distribution of X and Y is given by $f(x, y) = c(x^2 + y^2)$ for $x = -1, 0, 1, 3$ and $y = -1, 2, 3$. (i) Find the value of c . (ii) $P(x = 0, y \leq 2)$ (iii) $P(x \leq 1, y > 2)$ (iv) $P(x \geq 2 - y)$

Solution:

By data, $X = \{-1, 0, 1, 3\}$ and $Y = \{-1, 2, 3\}$

$$f(x, y) = c(x^2 + y^2)$$

The joint probability distribution of X and Y :

X \ Y	-1	2	3	$f(X)$
-1	$2c$	$5c$	$10c$	$17c$
0	c	$4c$	$9c$	$14c$
1	$2c$	$5c$	$10c$	$17c$
3	$10c$	$13c$	$18c$	$41c$
$g(Y)$	$15c$	$27c$	$47c$	$89c$

(i) **Find c :** $1 = \sum f(x, y) = 89c$

$$c = \frac{1}{89}$$

(ii) $x = 0, y = \{-1, 2\}$

$$P(x = 0, y \leq 2)$$

$$= P(0, -1) + P(0, 2)$$

$$= c + 4c = 5c$$

$$= 5/89$$

(iii) $x = \{-1, 0, 1\}, y = \{3\}$

$$P(x \leq 1, y > 2)$$

$$= P(-1, 3) + P(0, 3) + P(1, 3)$$

$$= 10c + 9c + 10c$$

$$= 29c = 29/89$$

Cont.

By data, $X = \{-1, 0, 1, 3\}$ and $Y = \{-1, 2, 3\}$

$$\begin{aligned} \text{(iv)} \quad P(x \geq 2 - y) &= P(x + y \geq 2) \\ &= P(-1, 3) + P(0, 2) + P(0, 3) + P(1, 2) + \\ &\quad P(1, 3) + P(3, -1) + P(3, 2) + P(3, 3) \\ &= 10c + 4c + 9c + 5c + 10c + 10c + 13c + 18c \\ &= 79c = 79/89 \end{aligned}$$

Practice Problems-set 2

- **Problem:1**

The joint probability mass function of (X, Y) is given by $P(x, y) = K(2x + 3y)$, $x = 0, 1, 2, y = 1, 2, 3$. Find i) the value of K , ii) The marginal probability function of X and Y , iii) $P(X = 2, Y \leq 2)$, iv) $P(X = 2)$.

Ans : i) $\frac{1}{72}$ ii) $P(X = 0) = \frac{18}{72}, P(X = 1) = \frac{24}{72}, P(X = 2) = \frac{30}{72},$
 $P(Y = 1) = \frac{15}{72}, P(Y = 2) = \frac{24}{72}, P(Y = 3) = \frac{33}{72}$ iii) $\frac{17}{72}$ iv) $\frac{30}{72}$.

- Show that X and Y are independent variables.

Problem: 2

Let X and Y have the following joint probability distribution

$X \backslash Y$	2	4
1	0.10	0.15
3	0.20	0.30
5	0.10	0.15

- Show that X and Y are independent variables.



Problem: 3

For the bivariate probability distribution of (X, Y) given below.

Find i) $P(X \leq 1)$, ii) $P(Y \leq 3)$, iii) $P(X \leq 1, Y \leq 3)$

Y X	1	2	3	4	5	6
0	0	0	$\frac{1}{32}$	$\frac{2}{32}$	$\frac{2}{32}$	$\frac{3}{32}$
1	$\frac{1}{16}$	$\frac{1}{16}$	$\frac{1}{8}$	$\frac{1}{8}$	$\frac{1}{8}$	$\frac{1}{8}$
2	$\frac{1}{32}$	$\frac{1}{32}$	$\frac{1}{64}$	$\frac{1}{64}$	0	$\frac{2}{64}$

Ans : i) $\frac{7}{8}$, ii) $\frac{23}{64}$ iii) $\frac{9}{32}$

Problem: 4 If $f(x, y) = e^{-(x+y)}$, $x \geq 0$, $y \geq 0$ is the joint probability density function of X and Y , find $P(X < 1)$. (Ans: 0.6321)

Problem: 5

The joint probability density function of X and Y given by

$$f(x, y) = \begin{cases} Kxy & 0 < x < 2, 0 < y < 2 \\ 0 & \text{otherwise} \end{cases}.$$

i) Find constant K ii) $P(\frac{1}{2} < X < \frac{3}{2}, 1 < Y < 2)$.

Ans : i) $\frac{1}{4}$ ii) $\frac{3}{8}$

Problem: 6

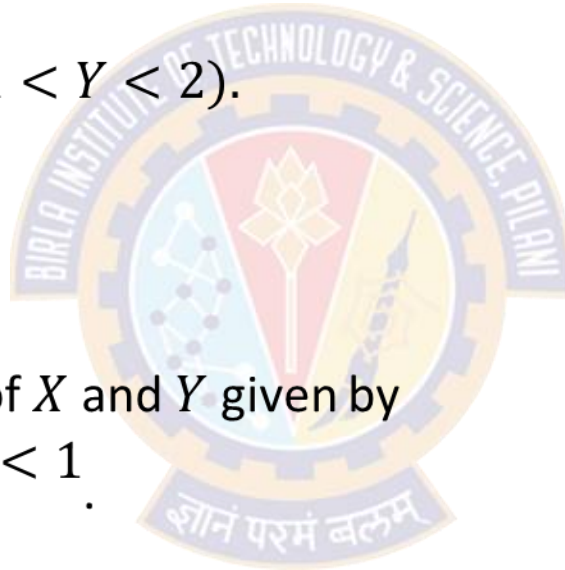
The joint probability density function of X and Y given by

$$f(x, y) = \begin{cases} x + y & 0 < x < 1, 0 < y < 1 \\ 0 & \text{otherwise} \end{cases}.$$

Find marginal probability function of X and Y . Also, check whether X and Y are independent or not.

Ans: Marginal prob function for X : $x + \frac{1}{2}$,

Marginal probability function of Y : $\frac{1}{2} + Y$; Not independent.





Thank You